FTEC5660 Homework 2 — Reflection

Four paradoxes of AI verification, practised on myself

This homework asked me to build a CV verification agent and then to fool verification agents in the same week. That structure is the point. Every paradox below is something I did to myself, not something I read about.

Paradox 1: the stronger checker is not the safer system. I hardened my verifier against shifted years, inflated titles and invented skills until it scored 7/7 on the public set. Then I wrote a CV that beats it without touching a single field it checks: every lie lives in a bullet description or the headline. The rule I added to suppress false positives — "descriptions are never evidence" — is exactly the door my attack walked through. Precision against a specification creates blindness outside the specification. Better checking of the checked does nothing for the unchecked.

Paradox 2: consistency is not truth. My verifier labels a CV "valid" when it agrees with LinkedIn. But LinkedIn is also self-authored. I am grading one piece of self-reported text against another and calling the result verification. In the mock world Facebook "confirms" LinkedIn because the data was generated that way; in the real world, two agreeing profiles are cheap to manufacture. KYC agents scale trust in self-reported data; they do not manufacture truth.

Paradox 3: the same model sits on both sides. Attacker and defender run the same model, `deepseek-v4-flash`. Every capability gain that helps it judge evidence also helps it obey the hidden white text in my PDF. Instruction-following is one skill serving two masters, and the model cannot tell whose instruction it is executing. My own defence in the end was to stop asking the model for a score at all — the score is computed in code from the list of discrepancies. That is an admission worth sitting with: the safest judge is the part of the system that cannot be persuaded because it cannot read.

Paradox 4: the complacency dividend. A tired human recruiter might well catch "CFA Charterholder" missing from the LinkedIn page they are staring at. My agent never will, because I forbade it from reading the headline. As verification gets cheaper, we audit less ourselves, and the lies that survive are precisely the ones tuned to the machine's blind spots. The fraudster does not need to fool everyone — only the cheapest checker in the pipeline.

What I take away. A verification agent is adversarial software whether we admit it or not; its specification is its attack surface. It should be tested the way this homework tested mine: write the attack before shipping the defence. And the deepest paradox of all: the harder my verifier gets, the more honest everyone must be — except the one person who studies exactly how honest it demands them to be.